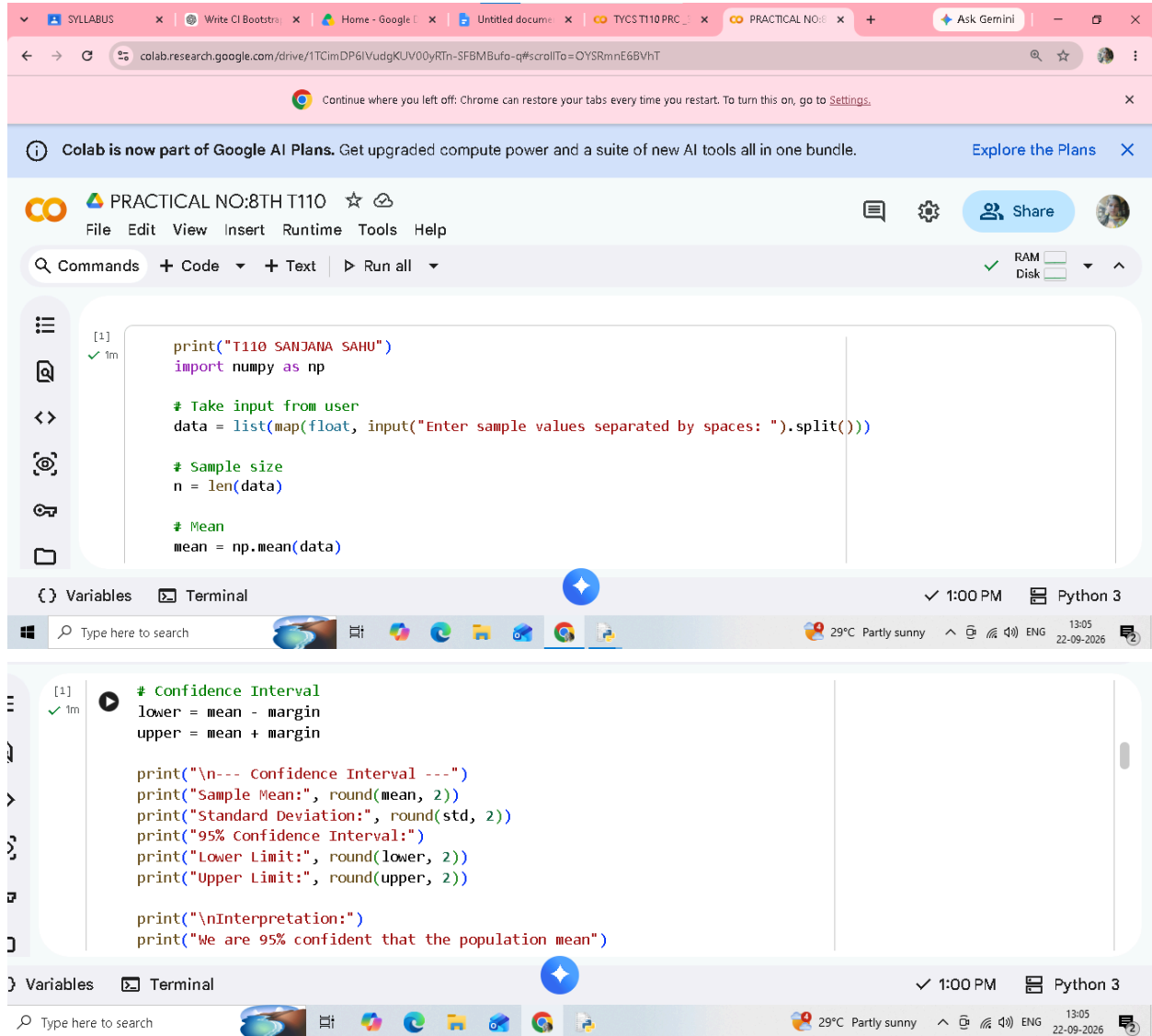


PRACTICAL NO:8TH

AIM:Confidence Interval Estimation:

- Calculate confidence intervals for population means or proportions using sample data and interpret the results in a business context.
- Apply bootstrapping techniques to estimate confidence intervals for non-parametric statistics.

a) Confidence Interval for Population Mean



The screenshot displays a Google Colab notebook titled "PRACTICAL NO:8TH T110". The notebook contains two code cells. The first cell defines variables for sample data and calculates the mean. The second cell calculates the confidence interval and prints the results.

```
[1] ✓ 1m
print("T110 SANJANA SAHU")
import numpy as np

# Take input from user
data = list(map(float, input("Enter sample values separated by spaces: ").split()))

# Sample size
n = len(data)

# Mean
mean = np.mean(data)

# Confidence Interval
lower = mean - margin
upper = mean + margin

print("\n--- Confidence Interval ---")
print("Sample Mean:", round(mean, 2))
print("Standard Deviation:", round(std, 2))
print("95% Confidence Interval:")
print("Lower Limit:", round(lower, 2))
print("Upper Limit:", round(upper, 2))

print("\nInterpretation:")
print("We are 95% confident that the population mean")
```

OUTPUT:

T110 SANJANA SAHU

PRACTICAL NO:8TH

```
T110 SANJANA SAHU
Enter sample values separated by spaces: 45 50 52 48 55 49 51 53 47 50

--- Confidence Interval ---
Sample Mean: 50.0
Standard Deviation: 2.94
95% Confidence Interval:
Lower Limit: 47.89
Upper Limit: 52.11

Interpretation:
We are 95% confident that the population mean
lies between 47.89 and 52.11
```

B) Bootstrapping

```
print("T110 SANJANA SAHU")
import numpy as np

# Take input from user
data = list(map(float, input("Enter sample values separated by spaces: ").split()))

data = np.array(data)

# Number of bootstrap samples
n_bootstrap = 10000
```

```
# Store bootstrap medians
bootstrap_medians = []

# Bootstrap sampling
for i in range(n_bootstrap):

    sample = np.random.choice(
        data,
        size=len(data),
        replace=True
    )
```

OUTPUT:

```
T110 SANJANA SAHU
Enter sample values separated by spaces: 45 50 52 48 55 49 51 53 47 50

--- Bootstrap Confidence Interval ---
Original Sample Median: 50.0
Number of Bootstrap Samples: 10000
95% Bootstrap Confidence Interval:
Lower Limit: 48.0
Upper Limit: 52.0

Interpretation:
The 95% bootstrap confidence interval for the
population median is between 48.0 and 52.0
```

T110 SANJANA SAHU